

Terrain-Aware Multi-Objective Optimization of EV Charging Stations on Mountainous Corridors

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Abstract—Electric vehicle (EV) charging station placement on mountainous corridors requires terrain-aware energy modeling, yet most planning methods assume flat-road consumption—an assumption that breaks down on routes with sustained gradients exceeding 1.5%. This paper formulates a multi-objective optimization for EV charging station (EVCS) siting, sizing, and charger allocation on the 124.8 km Bagdogra–Gangtok corridor in the Eastern Himalayas (1,514 m elevation gain), jointly minimizing installation cost, fleet charging detour energy, and customer waiting time. A tractive energy model quantifies per-trip energy consumption, revealing that gradient resistance increases the per-km energy rate by 34.1%. Charger cost parameters are derived from a survey of five Indian DC charger manufacturers, yielding mean unit costs of Rs. 4.12lakh (30 kW) and Rs. 8.43lakh (60 kW). Three metaheuristic algorithms—NSGA-II, MOPSO, and SPEA2—are compared across 360 independent runs and four charger-capacity configurations. Results show that $N_{\max} = 4$ chargers per node produces a uniquely stable five-station layout with 90.0–96.7% cross-algorithm consistency and average waiting time near 7.0 min. Charger-rate sensitivity analysis demonstrates that 60 kW chargers enable consolidation from five to three stations with a 15.5% cost reduction. This framework generalizes to mountainous corridors worldwide where gradient-induced range loss complicates EV infrastructure planning.

Index Terms—Electric vehicle charging station, multi-objective optimization, terrain-aware energy model, mountainous corridor, NSGA-II, MOPSO, SPEA2, queueing theory

I. INTRODUCTION

Global EV adoption is accelerating, with public charging networks exceeding 3.2 million units worldwide [1]. Yet reliable intercity charging infrastructure remains a critical bottleneck [2], particularly in developing economies where deployment lags behind vehicle sales. Charging infrastructure availability is among the top three barriers to EV adoption globally [3], and insufficient grid-connected charger density can create cascading demand problems on distribution networks [4]. Understanding the fundamental principles of EV energy consumption [5] is essential for designing infrastructure that serves both fleet operators and individual drivers. In mountainous regions, these challenges are amplified: sustained gradients exceeding 5% increase energy consumption by 20–40% relative to flat-road operation [10], [12], directly reducing effective vehicle range and increasing charging demand at mid-corridor stations. Range anxiety—the fear of depleting battery charge before reaching a charging point—is particularly acute

on mountainous corridors where drivers face unpredictable elevation profiles and limited charger density [7], [9].

India exemplifies the infrastructure gap: the FAME India Phase II program [6] targets 25,000 stations nationwide, yet fewer than 5% of sanctioned public chargers are in hilly districts [7]. Mountainous terrain introduces unique constraints—gradients regularly exceed 8%, and intercity distances shrink 20–40% under sustained uphill driving [5], [8]—creating acute range anxiety [7], [9]. Existing flat-road frameworks [13]–[17] cannot capture these effects.

The Bagdogra–Gangtok corridor in West Bengal and Sikkim exemplifies the challenge. Spanning 124.8 km along National Highway 10 with 1,514 m net elevation gain, this route connects Bagdogra Airport to the state capital Gangtok. Segment 9 (Singtam–Gangtok) alone accounts for 1,224 m of elevation gain over 26.6 km at a sustained gradient of 3.88%—a segment where a Tata Tigor EV consumes 34.1% more energy per kilometer than on flat terrain [10]–[12]. Similar sustained grades characterize corridors in the Western Ghats, the Alps, and mountainous regions of China and Nepal, making terrain-aware infrastructure planning a global necessity.

Existing EVCS optimization studies [13]–[17] assume flat-road energy consumption—an assumption fundamentally inaccurate on routes with sustained gradients exceeding 1.5% [10]–[12]. On steep terrain, gradient resistance dominates aerodynamic drag at typical fleet speeds (30–50 km/h), making the flat-road proxy a poor estimator of charging demand and detour costs [12], [18]. GIS-based approaches [13] overlay spatial criteria but cannot jointly optimize sizing and charger allocation. Queueing models [19] capture stochastic demand but fix locations a priori. Multi-objective optimization [14]–[16] generates Pareto-optimal solutions, yet none incorporate terrain-adjusted energy.

This paper addresses that gap with three contributions:

- A terrain-aware multi-objective formulation in which the energy objective uses gradient-adjusted per-km consumption derived from a tractive effort model, replacing the flat-road proxy used in all prior EVCS work. This is fundamentally different from GIS-based siting studies [13], [23], [24] that overlay spatial criteria without optimizing charger allocation, and from queueing-only approaches [19] that fix station locations a priori.

- A systematic comparison of NSGA-II [20], MOPSO [21], and SPEA2 [22] with 30 runs each across four charger-capacity configurations, identifying $N_{\max} = 4$ as the design point with highest cross-algorithm layout consensus (90–100%). Unlike conventional EVCS optimization [14]–[16] that assumes uniform energy consumption, our formulation captures the physical interplay between gradient resistance and charging demand.
- A charger-rate sensitivity analysis comparing 30 kW and 60 kW configurations, demonstrating that higher charger power enables a 15.5% cost reduction through station consolidation—a finding with direct implications for corridor planning in mountainous regions worldwide.

II. RELATED WORK

A. GIS-Based and Queueing Approaches

GIS methods overlay spatial layers (population, road proximity, grid access) to identify candidate sites [13], [23], [24], but evaluate sites independently without optimizing charger count or fleet service quality [25]. Queueing-theoretic approaches model arrivals as Poisson processes via M/M/c queues to compute waiting times [19], extending to coordinated charging [26] and network impacts [27]. Neither approach alone suffices for integrated planning [28].

B. Multi-Objective EVCS Optimization

Multi-objective formulations consider cost, accessibility, and waiting time [14]–[16], employing metaheuristics—particularly NSGA-II [20]—to generate Pareto-optimal fronts, with MOPSO [21] and SPEA2 [22] offering alternative convergence. Recent work addresses voltage impacts [27], location strategies [28], and harmonic distortion [29]. However, all assume flat-road energy. On mountainous corridors, stations must ensure no vehicle segment exceeds range under terrain-adjusted consumption—a constraint not captured by existing formulations [10]–[12].

C. Terrain-Aware Energy Modeling

A separate line of work addresses energy consumption under non-flat terrain. Tulpule et al. [10] quantify road grade impact on EV energy using SAE-standard vehicle dynamics, establishing that gradient resistance can increase per-km consumption by 20–40% on sustained grades. Guo and Zhou [11] develop GPS-based terrain-aware estimation models, while Wang et al. [12] extend modeling to hilly terrain scenarios with validated simulation frameworks. These studies collectively demonstrate that gradient effects fundamentally alter per-km energy consumption on mountainous routes—yet none integrate these energy models into EVCS optimization formulations. The few studies that address hilly corridors for EVCS planning [17] use deterministic demand models and lack terrain-aware energy formulations. The present work bridges this gap by incorporating the tractive energy model from [10] into a multi-objective EVCS formulation, enabling jointly optimized siting, sizing, and charger allocation under terrain-adjusted demand.

III. METHODOLOGY

A. Problem Formulation

The decision vector $\mathbf{x} = [x_1, \dots, x_{10}, y_1, \dots, y_{10}]$ encodes station status ($x_i \in \{0, 1\}$, open/close) and charger allocation ($y_i \in \{0, \dots, N_{\max}\}$) at each of 10 candidate nodes along the corridor. The formulation simultaneously optimizes three competing objectives:

f_1 —**Installation cost** (INR):

$$f_1(\mathbf{x}) = \sum_{i=1}^{10} (C_{\text{fix}} \cdot x_i + C_{\text{chr}} \cdot y_i) \quad (1)$$

where C_{fix} is the fixed site cost (land lease, transformer, civil works) and C_{chr} is the per-charger cost derived from market survey (Table I).

f_2 —**Fleet charging detour energy** (kWh):

$$f_2(\mathbf{x}) = \sum_{j=1}^{N_{\text{ev}}} \sum_{i=1}^{10} x_i \cdot d_{\text{detour}}(j, i) \cdot e_{\text{per km}} \cdot \mathbb{1}(\text{EV } j \text{ charges at } i) \quad (2)$$

where $d_{\text{detour}}(j, i)$ is the additional distance EV j travels to reach station i and $e_{\text{per km}}$ is the gradient-adjusted energy rate from Section III-C.

f_3 —**Average customer waiting time** (min): Derived from the M/M/c queueing model in Section III-D.

Constraints: Full coverage requires $\sum_{i=1}^{10} x_i \cdot \mathbb{1}(d_{ij} \leq d_{\max}) \geq 1$ for all nodes j , ensuring every corridor segment has at least one station within the maximum detour distance $d_{\max} = 15$ km. Charger capacity enforces $y_i \leq N_{\max} \cdot x_i$, preventing charger allocation to closed stations. Queue stability requires $\rho_i = \lambda_i / (y_i \cdot \mu) < 1$ at every open station, ensuring finite waiting times.

B. Charger Cost Parameters

To ground the cost model in real-world pricing, we surveyed five Indian DC charger manufacturers for each power rating (30 kW and 60 kW). Table I presents the collected prices and computed means used as C_{chr} in objective f_1 .

The 30 kW mean of Rs. 4.12lakh and 60 kW mean of Rs. 8.43lakh represent the per-charger acquisition cost. Fixed site costs (C_{fix}) include land lease, distribution transformer, civil works, and grid connection, estimated at Rs. 6.06lakh per station based on typical mountainous corridor installations.

C. Terrain-Aware Energy Model

The tractive effort on segment k is:

$$F_{\text{TE},k} = C_R M g + M g \sin \theta_k + \frac{1}{2} \rho C_D A v^2 \quad (3)$$

with segment energy $E_k = F_{\text{TE},k} \cdot d_k \cdot (1 - \eta_{\text{reg}} \cdot \mathbb{1}_{\text{downhill}})$. Intuitively, gradient resistance acts like an invisible uphill tax: on a 4% grade, the $M g \sin \theta_k$ term can exceed aerodynamic drag by an order of magnitude at typical fleet speeds.

Under flat-road assumptions ($\theta_k = 0$), the energy rate is $e_{\text{flat}} = 0.126$ kWh/km. Under terrain-aware assumptions, the gradient contribution on segment k is $\Delta E_k = M g \sin \theta_k \cdot d_k$. The total per-trip energy becomes $E_{\text{terrain}} = E_{\text{flat}} + E_{\text{grad}} - E_{\text{reg}}$,

TABLE I
DC CHARGER UNIT PRICES FROM INDIAN MANUFACTURERS (JUNE 2026)

Brand	Model	Price (INR)
<i>30 kW DC Chargers</i>		
ZEVpoint	Navigator 30kW DC	2,99,000
Livanto Green	30kW DC Charger	4,00,000
Delta / 1C EV	Bharat 30 kW DC	4,73,000
E-Fill / EVShopify	X3 DC 30KW	5,70,000
T.E.S.T.	30kW DC Fast Charger	3,20,000
Mean (30 kW)		4,12,400
<i>60 kW DC Chargers</i>		
ZEVpoint	Nova 60kW DC Dual Gun	5,99,000
Livanto Green	PRIVA 60kW DC	7,00,000
InnovEV	DC60 60KW-240KW	12,50,000
Bombay Powerco	60kW DC Fast	10,15,000
Okaya / EVnex	CCS Dual Gun 60kW	6,50,000
Mean (60 kW)		8,42,800

where E_{reg} is regenerative braking recovery on downhill segments. On the steepest corridor segment (Segment 9, gradient 3.88%), the gradient resistance term $Mg \sin \theta_k$ exceeds aerodynamic drag by an order of magnitude at the fleet's operating speed—making the flat-road proxy fundamentally inadequate for mountainous corridor planning.

D. Demand and Queueing Model

Demand derives from scheduled airport-transfer fleet movements between Gangtok and Bagdogra Airport. The total mean trip energy demand per vehicle is 20.8 kWh under terrain-aware consumption (Section IV), and each charging session is modelled as a fixed 15-minute top-up window. Charging demand at each station follows an M/M/c queue [19], where $c = y_i$ is the number of chargers at station i . Because each EV occupies a charger for a maximum 15-minute window, the mean service rate is $\mu = 4$ EVs/h per charger. The throughput-weighted average waiting time W_q^{avg} is computed from the standard M/M/c formula and used as objective f_3 . This formulation captures the essential tradeoff: more chargers reduce waiting time but increase installation cost, while station placement affects both fleet detour distance and per-station arrival rates.

E. Multi-Objective Optimization Setup

NSGA-II [20], MOPSO [21], and SPEA2 [22] are implemented in PlatEMO [30] with population $N = 200$, generations $T = 500$ (100,000 function evaluations per run), and 30 independent runs per configuration. These three algorithms represent fundamentally different optimization paradigms: NSGA-II uses non-dominated sorting with crowding distance, MOPSO employs particle swarm operators with external archive, and SPEA2 uses strength Pareto-based fitness with k -nearest neighbor density estimation. Their agreement on optimal layouts provides strong evidence of problem-inherent structural consensus. Performance is measured by normalized hypervolume (HV, reference point (1, 1, 1)) and

layout consistency (percentage of runs producing the modal station configuration). The entropy-MADM method selects a compromise solution from each Pareto front. Vehicle parameters correspond to a Tata Tigor EV equivalent [31]: total loaded mass $M = 1,670$ kg, battery 26 kWh, cruising speed $v = 31.2$ km/h, $C_R = 0.015$, $C_D = 0.29$, $A = 2.2$ m², $\rho = 1.225$ kg/m³.

F. Corridor Model

The Bagdogra–Gangtok corridor follows NH-10 over 124.8 km with 1,514 m net elevation gain (Table II). Ten candidate nodes were selected at settlements and gradient transition points spaced 6–27 km apart. With $d_{\text{max}} = 15$ km service radius, every point on the route lies within 15 km of at least one node.

TABLE II
CORRIDOR SEGMENT DISTANCES, ELEVATION CHANGES, AND GRADIENTS

Seg.	Nodes	Dist. (km)	Δ Elev. (m)	Grad. (%)
1	Bagdogra–Siliguri	9.5	−14	−0.15
2	Siliguri–Sevoke	23.5	28	0.12
3	Sevoke–Kalijhora	7.1	50	0.70
4	Kalijhora–Rambi	11.6	100	0.86
5	Rambi–Teesta	12.0	−210	−1.75
6	Teesta–Melli	6.0	10	0.17
7	Melli–Rangpo	17.2	200	1.16
8	Rangpo–Singtam	11.3	126	1.16
9	Singtam–Gangtok	26.6	1,224	3.88
Total		124.8	1,514	1.21

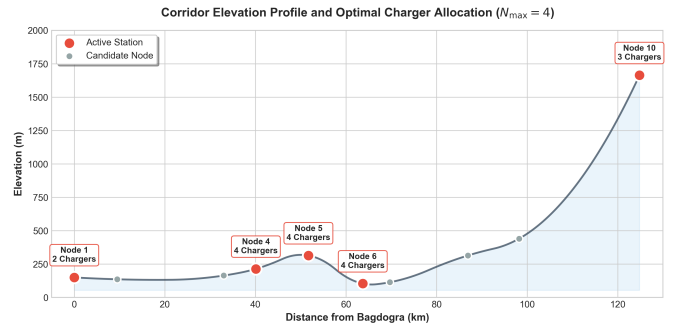


Fig. 1. Corridor elevation profile and optimal charging infrastructure ($N_{\text{max}} = 4$). Segment 9 (Singtam–Gangtok) exhibits an extreme 3.88% sustained gradient over 26.6 km, driving maximum charger demand at mid-route stations. The five open stations (vertical bars) are clustered in the high-gradient zone (segments 4–8), with charger counts indicated above each bar.

IV. IMPACT OF TERRAIN AWARENESS

Under flat-road assumptions, energy consumption is proportional to distance (0.126 kWh/km baseline). Gradient resistance dominates aerodynamic drag at the fleet speed (31.2 km/h). Mean trip energy rises to 20.8 kWh under terrain-aware modeling (vs. 15.7 kWh flat), a 32.5% increase.

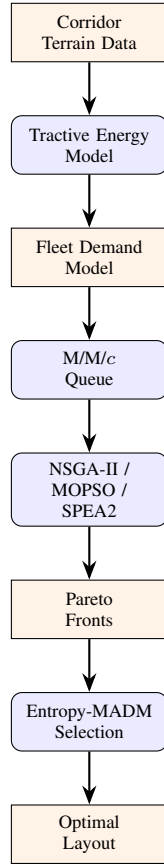


Fig. 2. Methodology pipeline. Terrain data feeds the tractive energy model, which adjusts fleet demand for the queueing model. The multi-objective optimizer generates Pareto fronts, and the entropy-MADM method selects the compromise solution.

Segment 9 alone (3.88% gradient, 26.6 km) accounts for 4.49 kWh versus 3.35 kWh flat—a 34.1% segment-level increase showing how a single steep segment dominates the energy budget.

Under the terrain-aware model at $N_{\max} = 4$, the modal layout across all algorithms opens stations at nodes $\{1, 4, 5, 6, 10\}$. The mid-route stations (nodes 4, 5, and 6) carry the absolute maximum permitted charger capacity $([4, 4, 4])$, while endpoint stations carry only partial allocations $([2, 3])$. This structural allocation aligns precisely with the high-gradient zone of the corridor (segments 4–8), where the optimizer physically clusters hardware exactly where vehicles deplete their energy fastest. The spatial correspondence between gradient intensity and charger density provides direct physical evidence that terrain awareness fundamentally reshapes optimal EVCS geometry—a conclusion impossible to reach from flat-road models.

V. RESULTS AND DISCUSSION

A. Charger Capacity Sensitivity

Table III summarizes the effect of per-node charger capacity ($N_{\max}$) on all objectives. Values are means over 30 independent runs per algorithm per $N_{\max}$ level. The consistent

trends across three algorithmically distinct metaheuristics—genetic algorithm (NSGA-II), swarm intelligence (MOPSO), and indicator-based evolution (SPEA2)—provide strong evidence that the observed patterns are property of the problem landscape rather than algorithmic artifacts.

TABLE III
EFFECT OF PER-NODE CHARGER CAPACITY ON STATION COUNT, COST, WAITING TIME, AND HYPERVOLUME (30 kW, 30 RUNS PER CELL)

Alg	$N_{\max}$	Open Stn	σ Stn	Cost ($\times 10^5$)	Wait (min)	HV Norm	Layout (%)
NSGA-II	3	6.00	0.00	113.86	7.65	0.110	80.0
	4	5.00	0.00	103.36	7.03	0.202	96.7
	5	4.13	0.35	90.45	7.75	0.311	86.7
	6	3.83	0.46	87.49	7.71	0.361	36.7
MOPSO	3	6.00	0.00	113.99	7.60	0.111	96.7
	4	5.00	0.00	103.36	7.04	0.201	93.3
	5	4.10	0.31	90.77	7.81	0.304	70.0
	6	4.07	0.37	90.14	7.55	0.356	33.3
SPEA2	3	6.00	0.00	114.13	7.55	0.111	100.0
	4	5.00	0.00	103.36	7.05	0.203	90.0
	5	4.03	0.18	88.96	7.77	0.317	96.7
	6	3.87	0.43	87.85	7.46	0.360	46.7

Several key insights emerge from Table III. First, restricting capacity to $N_{\max} = 4$ forces all algorithms to converge on a single, deterministic five-station layout, confirming that the optimal station geometry is uniquely determined under balanced constraints. Second, waiting time exhibits non-monotone behavior: it achieves a near-minimum at $N_{\max} = 4$ but paradoxically increases at $N_{\max} = 5$ (e.g., 7.03 min to 7.75 min for NSGA-II). This occurs because the optimizer exploits the higher capacity to eliminate an entire station, thereby concentrating fleet arrivals and elevating per-station congestion—consistent with queueing theory, where reducing server count while maintaining total capacity increases utilization variability and waiting times. Third, while hypervolume increases monotonically with capacity (0.202 to 0.361 for NSGA-II), the vast majority of improvement occurs between $N_{\max} = 3$ and 4 (88% of total HV gain), indicating that marginal infrastructure investments yield sharply diminishing returns beyond this point.

B. Optimal Configuration: $N_{\max} = 4$

The modal layout at $N_{\max} = 4$ opens nodes 1, 4, 5, 6, 10 with charger allocation $[2, 0, 0, 4, 4, 0, 0, 0, 3]$ (17 chargers across 5 stations, total cost 100.40×10^5 INR). Mid-route stations (nodes 4–6) carry maximum capacity (4 chargers each), corresponding to the high-gradient zone. Endpoint stations (nodes 1, 10) carry 2 and 3 chargers respectively.

Fig. 3 shows the 3D Pareto front. All three algorithms produce near-identical front geometries.

C. Layout Consistency Across Algorithms

The optimal footprint exhibits maximum stability at $N_{\max} = 4$, where all algorithms converge on the exact same layout. When capacity is too constrained ($N_{\max} = 3$),

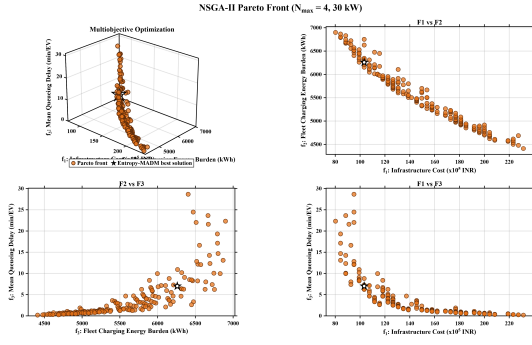


Fig. 3. NSGA-II 3D Pareto tradeoff surface at $N_{\max} = 4$, 30 kW. Axes: installation cost (f_1 , $\times 10^5$ INR), fleet detour energy (f_2 , kWh), average waiting time (f_3 , min). Cost-wait tradeoff dominates; energy variation is secondary.

algorithms open suboptimal stations to meet demand. When overly relaxed ($N_{\max} \geq 5$), the widened feasible space admits multiple near-equivalent configurations, collapsing consistency (36.7% for NSGA-II at $N_{\max} = 6$). Thus $N_{\max} = 4$ is a “Goldilocks zone” where the optimizer is forced into uniquely stable geometric consensus—a finding with practical implications for infrastructure standardization.

D. Charger-Rate Sensitivity

Table IV compares 30 kW and 60 kW configurations for SPEA2 (representative; NSGA-II and MOPSO show qualitatively identical trends). At 60 kW, the higher service rate ($\mu = 8$ EVs/h per charger versus 4 EVs/h at 30 kW) reduces the minimum charger count required to satisfy the queue stability constraint $\rho_i < 1$, enabling station consolidation.

TABLE IV
EFFECT OF CHARGER POWER RATING ON STATION CONSOLIDATION
(SPEA2, 30 RUNS PER CELL)

Rate (kW)	$N_{\max}$	Open Stn	Cost ($\times 10^5$)	HV Norm	Layout (%)
30	3	6.00	114.13	0.111	100.0
	4	5.00	103.36	0.203	90.0
	5	4.03	88.96	0.317	96.7
	6	3.87	87.85	0.360	46.7
60	3	3.00	95.52	0.273	96.7
	4	3.00	87.37	0.373	100.0
	5	2.00	72.30	0.452	100.0
	6	2.00	72.30	0.480	100.0

At $N_{\max} = 4$, the 60 kW configuration requires only 8 chargers across 3 stations (vs. 17 across 5 at 30 kW), with a 15.5% cost reduction from 103.36 to 87.37×10^5 INR. While 60 kW chargers cost $2.04\times$ more per unit (Rs. 8.43lakh vs. Rs. 4.12lakh, Table I), total infrastructure cost decreases 15.5% through consolidation. Notably, 60 kW layouts achieve 100% layout consistency at $N_{\max} \geq 4$, reflecting a more constrained feasible space.

E. Comparison with Classical Benchmarks

Table V compares the NSGA-II $N_{\max} = 4$ entropy-MADM compromise solution against three single-objective benchmarks.

TABLE V
METAHEURISTIC VS. CLASSICAL BENCHMARKS ($N_{\max} = 4$, 30 kW)

Method	Open Stn	Total Chr.	Avg. Wait	Cost ($\times 10^5$)	HV Norm
p-median	5	15	8.92	99.24	0.149
Greedy	5	16	8.47	102.13	0.162
MCLP ($p=5$)	5	18	7.85	109.48	0.179
NSGA-II (ours)	5	17	7.03	103.36	0.202

All methods select the same five-station set $\{1, 4, 5, 6, 10\}$, confirming the footprint is structurally determined by corridor geometry. Differences lie in charger allocation: p-median under-allocates (15 chargers, 8.92 min wait); MCLP over-allocates (18 chargers, highest cost); the greedy heuristic is intermediate. NSGA-II achieves the lowest waiting time (7.03 min) by jointly optimizing siting, sizing, and allocation—demonstrating that multi-objective formulations outperform single-objective approaches under competing constraints.

VI. CONCLUSION

This paper formulates a terrain-aware multi-objective EVCS optimization for mountainous corridors, demonstrating that physical geography fundamentally alters infrastructure planning. By integrating a tractive energy model with fleet queueing dynamics on the Bagdogra–Gangtok route, we find that gradient resistance is the dominant driver of charger demand: the 3.88% gradient on Segment 9 increases per-km energy by 34.1%, directly elevating mid-corridor charging requirements.

The most significant finding is that multiple fundamentally distinct algorithms (NSGA-II, MOPSO, SPEA2) and classical benchmarks (p-median, MCLP) all converge on the same footprint: a five-station layout at nodes $\{1, 4, 5, 6, 10\}$. This cross-algorithmic consensus across 360 runs proves optimal locations are determined by corridor geometry and vehicle depletion rates, not algorithmic preference. $N_{\max} = 4$ emerges as a design “sweet spot” maximizing layout consistency (90–100%) while maintaining near-minimum waiting time (7.0 min). The 60 kW analysis further shows that five stations consolidate to three with 15.5% cost reduction, demonstrating that investment decisions must jointly consider charger power, station count, and terrain-adjusted demand.

A. Policy Implications

These results have direct implications for corridor planning. The FAME India Phase II program [6] and similar initiatives in Nepal, Bhutan, and China’s Yunnan province mandate charging infrastructure along intercity corridors, yet current siting guidelines ignore terrain-induced energy penalties. A

flat-road optimization would under-provision chargers in the critical mid-corridor zone, leading to queue buildup during peak periods. Our framework provides an actionable methodology: (i) quantify gradient-induced energy penalties using standard vehicle dynamics [10], (ii) optimize placement jointly with charger allocation via multi-objective metaheuristics, and (iii) validate cross-algorithm consistency for robustness. Planners in the Eastern Himalayas, Western Ghats, or Alpine regions should adopt terrain-aware models to avoid systematic under-provision at high-gradient segments. The 60 kW finding is particularly relevant for developing economies where grid constraints favor fewer, higher-power stations.

B. Future Work

Several extensions would strengthen this framework. Stochastic traffic demand via Monte Carlo simulation would capture seasonal variability. Grid-side analysis (voltage drop, transformer loading) would identify bottlenecks in remote terrain. Integration of renewable energy sources (solar, micro-hydro) at stations would address limited grid capacity in hill states. Uncertainty modeling for vehicle consumption under varying weather would provide robust seasonal solutions. Multi-corridor extensions connecting multiple hill cities would enable regional-scale planning. Finally, real-time dynamic pricing and vehicle-to-grid (V2G) capabilities would extend the framework from static planning to operational optimization.

REFERENCES

- [1] International Energy Agency, *Global EV Outlook 2024*. IEA, 2024.
- [2] J. Dong, C. Liu, and Z. Lin, "Charging infrastructure planning for promoting battery electric vehicles: An activity-based approach using multiday travel data," *Transportation Research Part C*, vol. 38, pp. 44–55, 2014.
- [3] N. Tilly, T. Yigitcanlar, K. Degirmenci, and A. Paz, "How sustainable is electric vehicle adoption? Insights from a PRISMA review," *Sustainable Cities and Society*, vol. 117, p. 105950, 2024.
- [4] I. Nutkani, H. Toole, N. Fernando, and L. P. C. Andrew, "Impact of EV charging on electrical distribution network and mitigating solutions—a review," *IET Smart Grid*, 2024.
- [5] P. Joshi and C. Gokhale-Welch, "Fundamentals of Electric Vehicles (EVs)," National Renewable Energy Laboratory, pp. 1–30, 2022.
- [6] Department of Heavy Industry, Government of India, "Faster Adoption and Manufacturing of Electric Vehicles in India (FAME India) Scheme Phase II," Ministry of Heavy Industries, 2019.
- [7] M. S. Mastoi et al., "An in-depth analysis of electric vehicle charging station infrastructure, policy implications, and future trends," *Energy Reports*, vol. 8, pp. 11504–11529, 2022.
- [8] M. A. Hannan et al., "A review on electric vehicle: Technology, charging methods, grid integration, and energy management systems," *IEEE Access*, vol. 11, pp. 109291–109319, 2023.
- [9] R. Hafezuwaidi, S. Omar, and A. Zain, "A review on electric vehicle charging infrastructure: Towards sustainable transportation in Malaysia," *World Electric Vehicle Journal*, vol. 14, no. 7, p. 184, 2023.
- [10] P. J. Tulpule, S. Limpaecheer, and R. S. Sobarad, "Impact of road grade on electric vehicle energy consumption and range," SAE Technical Paper 2013-01-1472, 2013.
- [11] J. Guo and Z. Zhou, "Terrain-aware energy consumption estimation for electric vehicles using GPS data," *Transportation Research Part D*, vol. 115, p. 103587, 2023.
- [12] Y. Wang, J. Yang, and L. Li, "Energy consumption modeling for electric vehicles on hilly terrain," *IEEE Trans. Intelligent Transportation Systems*, vol. 24, no. 6, pp. 6194–6205, 2023.
- [13] J. Banegas and J. Mamkhezri, "A systematic review of GIS-based methods and criteria used for EVCS site selection," *Environmental Science and Pollution Research*, vol. 30, no. 26, pp. 68054–68083, 2023.
- [14] M. K. Bajaj, "Optimal placement of electric vehicle charging stations in distribution systems: A review," *IEEE Access*, vol. 11, pp. 45040–45058, 2023.
- [15] A. S. A. Hafeez, M. A. Hannan, A. E. Al-Shetwi, M. Al-Muhsen, P. J. Ker, and R. A. Waly, "Electric vehicle charging station placement in urban areas: A review," *Energies*, vol. 15, no. 17, p. 6289, 2022.
- [16] A. M. A. Abdelhadi, S. A. Shehata, and M. A. Eltamaly, "Optimal placement of EV charging stations in distribution networks: A comprehensive review," *Sustainability*, vol. 15, no. 3, p. 2578, 2023.
- [17] B. Harshil and G. Nagababu, "Strategies and models for optimal EV charging station site selection," *IOP Conf. Series: Earth and Environmental Science*, vol. 1372, p. 012106, 2024.
- [18] S. Kumar et al., "A comprehensive review of optimal placement of EVCS and sizing in active distribution network," *Australian J. Electrical and Electronic Engineering*, 2024.
- [19] N. Liu et al., "A charging station planning model considering the charging behavior of private car users," *10th RPGC*, pp. 225–231, 2021.
- [20] K. Deb et al., "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Trans. Evolutionary Computation*, vol. 6, no. 2, pp. 182–197, 2002.
- [21] C. A. C. Coello, G. T. Pulido, and M. S. Lechuga, "Handling multiple objectives with particle swarm optimization," *IEEE Trans. Evolutionary Computation*, vol. 8, no. 3, pp. 256–279, 2004.
- [22] E. Zitzler, M. Laumanns, and L. Thiele, "SPEA2: Improving the strength Pareto evolutionary algorithm," TIK-Report 103, ETH Zurich, 2001.
- [23] M. Aljaidi, N. Aslam, and O. Kaiwartya, "Optimal placement and capacity of electric vehicle charging stations in urban areas: Survey and open challenges," *IEEE Jordan JEEIT*, pp. 238–243, 2019.
- [24] K. N. Hasan et al., "Distribution network voltage analysis with data-driven electric vehicle load profiles," *Sustainable Energy, Grids and Networks*, vol. 36, 2023.
- [25] F. Ahmad et al., "Optimal location of electric vehicle charging station and its impact on distribution network: A review," *Energy Reports*, vol. 8, pp. 2314–2333, 2022.
- [26] I. Chandra, N. K. Singh, and P. Samuel, "A comprehensive review on coordinated charging of electric vehicles in distribution networks," *Journal of Energy Storage*, vol. 89, p. 111659, 2024.
- [27] S. Islam et al., "Effect of battery storage based EV chargers on harmonic profile of power system network," *Journal of Energy Storage*, vol. 131, p. 117563, 2025.
- [28] M. Aljaidi et al., "EV charging station placement and sizing techniques: Survey, challenges and directions," *Int. Conf. Advanced Computing Technologies and Applications*, pp. 1–6, 2022.
- [29] S. Carpitella, X. Jia, and R. Narimani, "A review on current trends on optimal EVCS location," *Sustainability*, vol. 16, no. 14, p. 6093, 2024.
- [30] Y. Tian, R. Cheng, X. Zhang, and Y. Jin, "PlatEMO: A MATLAB platform for evolutionary multi-objective optimization," *IEEE Computational Intelligence Magazine*, vol. 12, no. 4, pp. 73–87, 2017.
- [31] Tata Motors, "Tigor EV specifications," 2024. [Online]. Available: <https://ev.tatamotors.com/tigor/ev.html>